ADRI KARTADINATA

Electrical & Automation Engineer | Industrial Projects | Mining & Energy
Gresik, East Java | 085168600278 | adrikartadinata6@gmail.com | linkedin.com/in/adri-kartadinata
Ahli K3 Umum — Kemenaker RI (2026) | *Senior Mechatronics Technician — BNSP (2025)*

PROFESSIONAL SUMMARY

Results-driven Automation Engineering graduate with expertise in PLC programming, Industrial IoT, instrumentation, and industrial automation. Experienced in furnace automation, mechanical drafting, and site supervision for high-value projects with PT Freeport Indonesia. Certified AK3 Umum (Kemenaker RI) professional with strong capabilities in industrial control systems, project execution, K3/HSE compliance, procurement, and technical documentation. Proven ability to manage multidisciplinary engineering projects from tendering through commissioning, while driving continuous improvement initiatives and maintaining high safety standards.

PROFESSIONAL EXPERIENCE

PT Aldzama | Gresik, East Java
Junior Engineer – Instrumentation and Robotics | *October 2025 – Present*

Awards & Recognition

- Honored with the Best Idea Prize (Kaizen Improvement Category) and Most Active Prize (Improvement Category) for outstanding contributions to process optimization and workplace innovation.

Project Engineer — PTFI Copper Chip Milling System | *December 2025 – Present*

- Engineered the complete tender documentation package — method statements, technical specifications, equipment BOQ, floor plan layouts, project schedule, and K3/HSE plan — securing contract award for the Copper Chip Milling System project at PT Freeport Indonesia.
- Led the formal technical client presentation covering project scope, commissioning methodology, K3 compliance framework, and equipment selection rationale.
- Coordinated multi-vendor equipment procurement and a Gantt-based master project schedule, aligning delivery timelines with the critical path to prevent schedule slippage.
- Authored K3/HSE documentation to full PTFI safety management system standards: JSA, HIRADC, Permit-to-Work, Method Statements, and Emergency Response Plans.

Furnace Operation & Burner Control Engineer | *December 2025 – Present*

- Engineered and commissioned Omron PLC-based furnace control programs for LPG burner systems, improving temperature regulation accuracy across heat-soaking zones and reducing the need for manual operator intervention during hold periods.
- Designed electrical wiring layouts for furnace panel equipment to PUIL/IEC 60364 standards and monitored LPG supply pressure across all burner lines on every production shift.
- Executed castable refractory curing cycles with precision temperature staging to prevent thermal-shock-induced cracking across completed components.

Project Supervisor — Kiln Demolition, PT Hensheng New Energy Material Indonesia | *May 2026*

- Spearheaded on-site supervision of kiln demolition workflows, directing field technicians and subcontractor teams to deliver the full scope on schedule with a verified zero-LTI safety record.
- Enforced K3/HSE compliance protocols at every phase in full conformance with Kemenaker RI regulations, maintaining zero reportable incidents from mobilization through demobilization.
- Engineered a structured daily reporting framework integrating work-progress metrics, material consumption data, and safety observations for real-time management visibility.

Refractory Lining Production Lead — Launder Cover, PTFI | *March 2026 – May 2026*

- Led the on-site fabrication team for specialized launder cover refractory systems for PT Freeport Indonesia.
- Audited daily raw material usage, enforced quality control parameters, and compiled client progress tracking files.

Mechanical Drafter | *January 2026 – April 2026*

- Draft 2D technical specifications and assembly drawings using AutoCAD and SolidWorks.

R&D Electrical Engineer | *November 2025 – January 2026*

- Conducted feasibility studies for industrial automation prototypes and assembled prototype control panels integrating contactors, relay logic modules, inverters, and PLC I/O wiring to IEC/PUIL standards.
- Supported instrumentation design reviews, technical documentation, and field installation support for control and hydraulic systems.

Administration Intern | *July 2023 – August 2023*

- Developed and standardized refinery operational SOPs for PT Freeport Indonesia.

PT Petrokimia Gresik | Gresik, East Java
Maintenance Engineer Intern — Instrumentation & Control Systems | *March 2024 – September 2024*

- Overhauled bagging system maintenance procedures covering weigher feeder calibration, dump door repair, and conveyor drive troubleshooting, consistently restoring units to operational specification.
- Commissioned control valve maintenance routines — pneumatic actuator adjustment, positioner installation, and post-calibration verification — reducing process deviation incidents across the monitored period.
- Gained hands-on exposure to Siemens S7-300 modular PLC systems with field-replaceable, plug-in I/O modules, supporting I/O module diagnostics and control panel troubleshooting alongside the maintenance team.
- Diagnosed and resolved faults in proximity sensors, pull cord switches, solenoid valves, and double-acting cylinders using P&ID interpretation and wiring panel analysis, reducing recurrence of failures.
- Led the field induction program for new intern cohorts; recognized by the Maintenance Division Head for training quality and leadership effectiveness.

ACADEMIC & PROJECT EXPERIENCE

Politeknik Manufaktur Bandung (Polman Bandung) | Bandung, West Java
Mechatronics Engineering Coursework & Laboratory Practicums | *2021 – 2025*

- Programmed and debugged ladder logic and relay-based control systems for sequential control, timer/counter logic, and interlocking automation exercises in the Teknik Kendali (Control Engineering) and Otomasi Industri I & II (Industrial Automation I & II) courses.
- Designed and tested sensor/actuator interfacing circuits, including pneumatic actuator control logic, in the Sensor dan Aktuator (Sensors & Actuators) and Instrumentasi dan Pengukuran (Instrumentation & Measurement) practicums.

- Designed and simulated electric motor control circuits — starting, reversing, and interlocking logic — in the Elektronika Daya (Power Electronics) and Otomasi Industri coursework, translating control requirements into wiring diagrams and I/O schematics.
- Completed hands-on laboratory practicums across Mikrokontroler dan Antarmuka (Microcontroller & Interfacing), Sistem Mekatronika (Mechatronic Systems), and Pengembangan Produk Mekatronika (Mechatronic Product Development), building the multidisciplinary automation and control foundation later applied professionally at PT Aldzama and PT Petrokimia Gresik.

Final Year Thesis Project — Underwater Object Detection Sonar System | *January 2025 – September 2025*

- Designed and built a six-transducer piezoelectric sonar array with Arduino Uno-based control and real-time signal acquisition logic for object detection.
- Developed an integrated Python-based real-time data visualization interface to monitor and interpret sensor output.

TECHNICAL SKILLS

Automation & Controls: PLC — Omron CX-One, Mitsubishi GX Works, Siemens (Ladder Logic, FBD, Structured Text); HMI — NB-Designer (Omron); SCADA — Node-RED, MQTT; LabVIEW; MATLAB/Simulink (PID, Fuzzy Logic)

Electrical & Panel Design: AutoCAD Electrical (Wiring Diagrams, Panel Layout, SLD); Relay Logic; Power Distribution; PUIL / IEC 60364; P&ID Development

Embedded Systems & IoT: ESP32, Arduino, STM32; Modbus TCP, MQTT, OTA Firmware; PCB Design — Altium Designer, Eagle, Proteus

Mechanical & Drafting: SolidWorks (3D Modeling & Assembly), AutoCAD 2D/3D, Technical Drawing, BOM Generation, Fluidsim (Pneumatics / Hydraulics)

Instrumentation & Maintenance: Pressure, Flow, Level & Temperature Transmitters; Calibration; FAT/SAT/Loop Check; Preventive & Predictive Maintenance; Conveyor & Pneumatic Systems

HSE & Project Engineering: JSA, HIRADC, Method Statement, Permit-to-Work, Emergency Response Plan; BOQ; Gantt Scheduling; Procurement Coordination

Programming: Python, C/C++ (Embedded), Ladder Logic

Licenses: Valid Indonesian Driving Licenses — SIM A & SIM C

EDUCATION

Politeknik Manufaktur Bandung (Polman Bandung) | Bandung, West Java

Bachelor of Applied Engineering in Mechatronics Engineering (S.Tr.T.) | *August 2021 – September 2025*

GPA: 3.28 / 4.00 | Predicate: Sangat Memuaskan | Accreditation: BAN-PT "Baik Sekali" (Excellent)

- Thesis Project: Developed an underwater object detection sonar system using six piezoelectric transducers controlled via Arduino Uno, featuring an integrated Python-based real-time data visualization UI.
- Relevant Coursework: Teknik Kendali (Control Engineering), Otomasi Industri I & II (Industrial Automation I & II), Sensor dan Aktuator (Sensors & Actuators), Mikrokontroler dan Antarmuka (Microcontroller & Interfacing), Sistem Mekatronika (Mechatronic Systems), Elektronika Daya (Power Electronics), Pemodelan Sistem (System Modeling), Kendali Cerdas (Intelligent Control), Instrumentasi dan Pengukuran (Instrumentation & Measurement), Digital dan Mikroprosesor (Digital Systems & Microprocessors), Pengembangan Produk Mekatronika (Mechatronic Product Development), Gambar Teknik/CAD (Technical Drawing/CAD)

SMAN 1 Gresik | Gresik, East Java

Science & Mathematics | *GPA: 87.98 / 100* | *July 2018 – May 2021*

CERTIFICATIONS

- General Occupational Health & Safety Expert (Ahli K3 Umum) — Kemenaker RI — Ministry of Manpower (2026)
- Senior Mechatronics Technician (Teknisi Senior Mekatronika) — BNSP — National Professional Certification Agency (2025)
- Letter of Completion — Industrial Maintenance Engineering — PT Petrokimia Gresik (2024)
- Basic Leadership Training — Politeknik Manufaktur Bandung (2021)

ORGANIZATIONAL EXPERIENCE

- Group Leader — Community Service Program (KKN) | Politeknik Manufaktur Bandung (October 2024 – November 2024)
- President — Student Choir | Politeknik Manufaktur Bandung (April 2023 – December 2023)
- Deputy Head — Administrative & Secretarial Bureau (HIMAMO) | Politeknik Manufaktur Bandung (2021 – 2025)

LANGUAGES

- Indonesian — Native proficiency
- Javanese — Native proficiency
- Sundanese — Native proficiency
- English — Professional working proficiency

Nomor Ijasah Nasional : 00500121312202510001 9

No. Seri : 022/S.Tr./I/04/210/1734/2025

POLITEKNIK MANUFAKTUR BANDUNG

(Nomor Pokok Perguruan Tinggi : 005001)

menyatakan bahwa

Adri Kartadinata, S.Tr.T.

NIM : 221341001

Lahir di **Gresik** tanggal **4 Agustus 2003**

telah berhasil menyelesaikan Program Pendidikan **Sarjana Terapan**

pada tanggal 1 September 2025

Program Studi **Teknologi Rekayasa Mekatronika (Nomor Pokok Program Studi : 21312)**

Terakreditasi BAN-PT (Akreditasi Prodi **"Baik Sekali"** Surat Keputusan Nomor 0075/SK/LAM Teknik/VST/IV/2025, Akreditasi Institusi **"Baik Sekali"** Surat Keputusan Nomor 523/SK/BAN-PT/Akred/PT/VI/2021)

sehingga kepadanya diberikan ijazah Sarjana Terapan Politeknik dan berhak untuk menggunakan gelar

Sarjana Terapan Teknik (S.Tr.T.)

Bandung, **27 September 2025**

Ketua Jurusan
Teknik Otomasi Manufaktur dan Mekatronika,

Ridwan, S.S.T., M.Eng.
NIP. 197806122001121002



Direktur,

Darma Firmansyah Undayat, S.S.T., M.T.
NIP. 197602132003121003

Ijazah ini diterbitkan berdasarkan Surat Keputusan Direktur Nomor :
0556/PL11/KD/2025

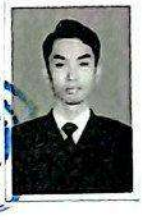
Izin Penyelenggaraan Perguruan Tinggi Nomor :
257/O/1998 tanggal 21 Oktober 1998

POLITEKNIK MANUFAKTUR BANDUNG

(POLITEKNIK MEKANIK SWISS - ITB)

TRANSKRIP AKADEMIK SARJANA TERAPAN
Program Studi: Teknologi Rekayasa Mekatronika
Jurusan: Teknik Otomasi Manufaktur dan Mekatronika

Ijin Penyelenggaraan Perguruan Tinggi Nomor: 257/O/1998 Tanggal 21 Oktober 1998



NPPT/NPPS	**: 005001/21312**
No. Transkrip	**: 022/S.Tr./T/04/210/1734/2025**
Nama	**: Adri Kartadinata**
NIM	**: 221341001**
Tempat/Tgl Lahir	**: Gresik, 4 Agustus 2003**
Tanggal Kelulusan	**: 1 September 2025**
Nomor Ijazah Nasional	**: 00500121312202510019**

Mata Kuliah Semester 1	sks	Nilai
Matematika dan Sains Alam		
Fisika 1	2	B
Matematika 1	2	AB
Kimia Dasar	2	AB
Sains Teknik		
Logika dan Pemrograman Komputer	2	AB
Teknologi Mekatronika		
Analisis Rangkaian Elektrik	2	C
Praktik Analisis Rangkaian Elektrik	2	B
Gambar Teknik Elektrik	2	AB
Proses Manufaktur Mekatronika	2	B
Metrologi & Statistika Pengukuran	2	C
Jumlah SKS		**18**
NK:4	**IP**	**2.96**

Mata Kuliah Semester 2	sks	Nilai
Sosial & Humaniora		
Olah Raga	2	AB
Matematika & Sains Alam		
Matematika 2	2	C
Fisika Terapan Mekatronika	2	C
Mekanika Teknik	2	B
Teknologi Manufaktur		
Gambar Teknik Mesin	3	B
Proses Manufaktur 1	3	B
Teknologi Mekatronika		
Elektronika	2	C
Praktik Elektronika	2	BC
Pemrograman Teknik	2	AB
Jumlah SKS		**20**
NK:4	**IP**	**2.81**

Mata Kuliah Semester 3	sks	Nilai
Matematika dan Sains Alam		
Persamaan Diferensial	2	C
Sains Teknik		
Material Teknik	2	B
Konsep Teknologi	2	AB
Teknologi Otomasi dan Mekatronika		
Digital dan Mikroprosesor	2	BC
Teknik Kendali	2	BC
Sensor dan Aktuator	2	B
Praktik Digital dan Mikroprosesor	2	A
Praktik Teknik Kendali	2	AB
Praktik Sensor dan Aktuator	2	A
Teknologi Manufaktur		
Praktik Proses Manufaktur 2	2	B
Jumlah SKS		**20**
NK:3	**IP**	**3.19**

Mata Kuliah Semester 4	sks	Nilai
Sosisal dan Humaniora		
Bahasa Indonesia	2	B
Matematika dan Sains Alam		
Aljabar Linier	2	BC
Sains Teknik		
Mekanika Teknik 3	2	B
Teknologi Mekatronika		
Teknik Produksi	2	B
Pemodelan Sistem	2	BC
Mikrokontroler	2	AB
Praktik Mikrokontroler dan Antarmuka	2	B
Praktik Elektronika Otomasi Industri	2	B
Praktek Pemrograman Labview	2	AB
Jumlah SKS		**18**
NK:2	**IP**	**3.04**

Mata Kuliah Semester 5	sks	Nilai
Sosial dan Humaniora		
Bahasa Inggris	2	A
Kesehatan, Keselamatan, dan Lingkungan	2	AB
Teknologi Mekatronika		
Statistika	2	A
Sistem Mekatronika	2	A
Elektronika Daya	2	B
Praktik Elektronika Daya	2	AB
Praktik Elektronika Otomasi Industri	2	AB
Praktik Pengembangan Produk Mekatronika	2	B
Komunikasi Data	2	AB
Manajemen Teknik		
Manajemen Industri	2	A
Jumlah SKS		**20**
NK:4	**IP**	**3.56**

Mata Kuliah Semester 6	sks	Nilai
Sosial dan Humaniora		
Pancasila dan Kewarganegaraan	2	A
Manajemen Teknik		
Manajemen Proyek	2	AB
Teknologi Mekatronika		
Praktik Produksi	7	A
Praktik Perawatan dan Perbaikan	2	A
Praktik Pengelolaan Alat dan Bahan	2	A
Praktik Perencanaan dan Pengendalian Produksi	2	A
Praktik Supervisi	2	A
Jumlah SKS		**19**
NK:4	**IP**	**3.88**

Mata Kuliah Semester 7	sks	Nilai
Sosial dan Humaniora		
Pancasila dan Kewarganegaraan	2	A
Manajemen Teknik		
Manajemen Mutu	2	AB
Teknologi Mekatronika		
Tugas Akhir	2	C
Praktik Supervisi	1	A
Pengabdian pada Masyarakat (KKN)	4	A
Perencanaan dan Pengendalian Produksi	3	A
Praktik Perawatan dan Perbaikan	1	A
Praktik Pengelolaan Alat dan Bahan	1	A
Tugsas Akhir (praktik)	1	C
Jumlah SKS		**17**
NK:4	**IP**	**3.49**

Mata Kuliah Semester 8	sks	Nilai
Sosial dan Humaniora		
Agama	2	AB
Ekonomi Teknik	2	B
Komunikasi	2	AB
Teknologi Mekatronika		
Sistem Kendali Cerdas	2	B
Robotika	2	C
Kapita Selekta Mekatronika	2	A
Praktik Kendali Cerdas	2	AB
Tugas Akhir (Praktik)	4	AB
Jumlah SKS		**18**
NK:4	**IP**	**3.33**

Nilai Kelakuan Kumulatif (NKK):29	**Indeks Prestasi Kumulatif (IPK): 3.28**	**Yudisium:Sangat Memuaskan**

Direktur,

Darma Firmansyah Undayat, S.S.T., M.T.
NIP.197602132003121003

Bandung, 27 September 2025
Ketua Jurusan Teknik Otomasi Manufaktur dan Mekatronika

Ridwan, S.S.T., M. Eng.
NIP.197806122001121002

BADAN NASIONAL
SERTIFIKASI PROFESI
INDONESIAN PROFESSIONAL CERTIFICATION AUTHORITY

SERTIFIKAT KOMPETENSI
CERTIFICATE OF COMPETENCE

No. 78411 8171 5 0001172 2025

Dengan ini menyatakan bahwa,
This is to ceritify that,

Adri Kartadinata

No. Reg. MEK 1363 00113 2025

telah kompeten pada bidang :
is competent in the area of :

Mekatronika
Mechatronics

dengan Kualifikasi/Kompetensi :
with Qualification/Competency :

Teknisi Senior Mekatronika
Senior Mechatronics Technician

Sertifikat ini berlaku untuk : 3 (Tiga) Tahun
This certificate is valid for : 3 (Three) Year

Bandung, 13 Agustus 2025

Atas Nama Badan Nasional Sertifikasi Profesi
On Behalf of the National Professional Certification Agency
Lembaga Sertifikasi Profesi Polman Bandung
Polman Bandung Professional Certification Body

LSP

Dr. Aris Budiyarto, S.T., M.T.

BNSP

PT. SH Indonesia
Pelatihan, Sertifikasi, Konsultan dan Analisa Lingkungan
❖ Permata Safira Regency Blok A4 No.5 Lidah Kulon, Lakarsantri, Surabaya
❖ Wisata Bukit Mas London Park Avenue AA2 / 17 Lidah Wetan, Lakarsantri, Surabaya
www.shindonesia.co.id

SURAT KETERANGAN

165/SHI.ADM/III/2026

Yang bertanda tangan di bawah ini menerangkan bahwa :

Nama : ADRI KARTADINATA
Perusahaan : PT. ALDZAMA

Telah dinyatakan **LULUS** dalam Pembinaan & Lisensi K3

AHLI K3 UMUM

Sertifikasi Kemnaker RI

Yang telah diselenggarakan oleh PT. SH Indonesia secara online pada tanggal **18 Februari - 5 Maret 2026** di Surabaya.

Surat Keterangan ini diberikan untuk dapat dipergunakan sebagaimana mestinya, sebagai pengganti sementara Sertifikat, SKP, dan Kartu Kewenangan yang masih dalam proses penerbitan oleh Kementerian Ketenagakerjaan RI.

Surat Keterangan ini berlaku sampai dengan diterbitkannya Sertifikat, SKP, dan Kartu Kewenangan resmi.

Demikian Surat Keterangan ini dibuat oleh Tim Pembinaan & Lisensi PT. SH Indonesia.

Surabaya, 12 Maret 2026
Hormat kami

PT SH Indonesia
Endang Sri Handayani, SP., MM.
Direktur